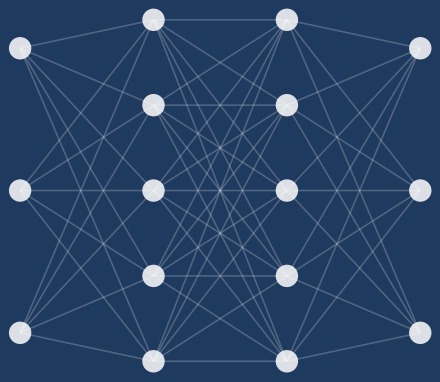
DEEP LEARNING FOUNDATIONS | PRACTICAL LABS

Practical Lab 3

Adaptive Optimizers for ANN and MLP

Dataset: MNIST handwritten digits
Framework: TensorFlow / Keras



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PROGRAM	Artificial Intelligence | 4th Semester
COURSE	Deep Learning Foundations
LAB	Lab 3 - Adaptive Optimizers for ANN and MLP (MNIST)
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Executed code (Google Colab notebook)

https://colab.research.google.com/github/24pwai0015-max/ann-and-deep-learning/blob/main/labs/Lab3_Adaptive_Optimizers_ANN_MLP.ipynb

Click the link to open the notebook in Colab (Runtime > Run all).

1. Introduction

Training a neural network means repeatedly nudging its weights in the direction that reduces the loss. The **optimizer** is the rule that turns the gradient into that weight update. This lab keeps the data, the architecture, the starting weights, the number of epochs and the batch size fixed, and swaps *only* the optimizer, so that any difference in the results can be attributed to the update rule.

Two fully connected networks are compared on a subset of the MNIST handwritten-digit dataset: a shallow **ANN** with one hidden layer (Flatten → Dense(64) → Dense(10)) and a deeper **MLP** with two hidden layers (Flatten → Dense(128) → Dense(64) → Dense(10)). Both are feed-forward neural networks; “ANN” simply denotes the smaller, one-hidden-layer version in this lab.

Optimizers and settings used

Optimizer	Main idea	Learning rate
SGD	Plain gradient step with one fixed rate (baseline)	0.01
Adam	Moving averages of the gradient and of its square	0.001
RMSprop	Divides each step by a running average of squared gradients	0.001
Adadelta	Scales updates using running averages of gradients and past updates	1.0
Adagrad	Accumulates all past squared gradients; shrinks steps over time	0.01

2. Experimental setup

Item	Setting
Dataset	MNIST (28 x 28 grayscale digits, 10 classes)
Subset sizes	Training 3000 | Validation 600 | Test 1200 (fixed random subsets, seed 42)
Architecture 1 (ANN)	Flatten → Dense(64, ReLU) → Dense(10, softmax)
Architecture 2 (MLP)	Flatten → Dense(128, ReLU) → Dense(64, ReLU) → Dense(10, softmax)
Fixed training settings	EPOCHS = 4, BATCH_SIZE = 64, loss = sparse categorical cross-entropy, metric = accuracy
Initial weights	Saved once per architecture and restored with `set_weights()` before every optimizer run
Only change between runs	The optimizer constructor (five optimizers, five learning rates)

3. Source code

The complete program is shown below with an explanatory comment on every statement. It is the same code that is executed in the Colab notebook **Lab3_Adaptive_Optimizers_ANN_MLP.ipynb**: open in Google Colab.

```
# LAB 3: Compare SGD, Adam, RMSprop, Adadelta and Adagrad on an ANN and an MLP.
# Run in Google Colab or locally after: pip install tensorflow numpy matplotlib
# Dataset: MNIST handwritten digits, downloaded automatically by Keras on first use.

import os  # Access an environment setting that reduces TensorFlow log noise.
os.environ["TF_CPP_MIN_LOG_LEVEL"] = "2"  # Show fewer TensorFlow informational messages.
import csv  # Export an easy-to-read results table without extra packages.
import time  # Measure how long each training experiment takes.
from pathlib import Path  # Create a folder for result files on any operating system.
import numpy as np  # Select the same images for every experiment.
import matplotlib.pyplot as plt  # Create accuracy comparison graphs.
import tensorflow as tf  # Build, train and test neural networks.

```

```
SEED = 42  # Use a repeatable random seed for classroom comparisons.
TRAIN_N = 3000  # Limit training images so the lab runs quickly on any machine.
VAL_N = 600  # Keep separate validation images for model selection.
TEST_N = 1200  # Keep a separate test subset for the final evaluation.
EPOCHS = 4  # Use the same number of training epochs for every optimizer.
BATCH_SIZE = 64  # Keep the mini-batch size fixed while comparing optimizers.
OUT = Path("lab3_optimizer_results")  # Name the folder that will hold the result files.
OUT.mkdir(parents=True, exist_ok=True)  # Create the folder if it does not exist yet.
tf.keras.utils.set_random_seed(SEED)  # Set the Python, NumPy and TensorFlow seeds.

(x_all, y_all), (x_test_all, y_test_all) = tf.keras.datasets.mnist.load_data()  # Load public MNIST.
rng = np.random.default_rng(SEED)  # Create a deterministic NumPy random generator.
train_order = rng.permutation(len(x_all))  # Shuffle the original training indices one time.
test_order = rng.permutation(len(x_test_all))  # Shuffle the separate test indices one time.
selected = train_order[:TRAIN_N + VAL_N]  # Select non-overlapping train and validation indices.
x_train = x_all[selected[:TRAIN_N]].astype("float32") / 255.0  # Scale training pixels to 0..1.
y_train = y_all[selected[:TRAIN_N]]  # Select training labels using the matching indices.
x_val = x_all[selected[TRAIN_N:]].astype("float32") / 255.0  # Scale validation pixels to 0..1.
y_val = y_all[selected[TRAIN_N:]]  # Select the validation labels.
x_test = x_test_all[test_order[:TEST_N]].astype("float32") / 255.0  # Scale held-out test pixels.
y_test = y_test_all[test_order[:TEST_N]]  # Select the matching test labels.
print("Train:", x_train.shape, "| Validation:", x_val.shape, "| Test:", x_test.shape)  # Confirm sizes.

# A shallow ANN is also technically an MLP; here "ANN" means one hidden layer.
def build_ann():  # Define the smaller one-hidden-layer network.
    return tf.keras.Sequential([  # Construct a Keras network with ordered layers.
        tf.keras.layers.Input(shape=(28, 28)),  # Receive one 28-by-28 image.
        tf.keras.layers.Flatten(),  # Convert the image into 784 numeric features.
        tf.keras.layers.Dense(64, activation="relu"),  # Learn 64 hidden features.
        tf.keras.layers.Dense(10, activation="softmax"),  # Predict the 10 digit classes.
    ])  # End the ANN model definition.

# In this lab, "MLP" means a deeper multilayer perceptron with two hidden layers.
def build_mlp():  # Define a two-hidden-layer multilayer perceptron.
    return tf.keras.Sequential([  # Construct the second Keras network.
        tf.keras.layers.Input(shape=(28, 28)),  # Receive a grayscale image.
        tf.keras.layers.Flatten(),  # Flatten 28-by-28 pixels into 784 features.
        tf.keras.layers.Dense(128, activation="relu"),  # Learn 128 first-layer features.
        tf.keras.layers.Dense(64, activation="relu"),  # Combine them in 64 deeper units.
        tf.keras.layers.Dense(10, activation="softmax"),  # Produce class probabilities.
    ])  # End the MLP model definition.

# These are illustrative starting learning rates, not necessarily the Keras defaults.
OPTIMIZERS = [  # Define the ONLY line of the experiment that changes between runs.
    ("SGD", lambda: tf.keras.optimizers.SGD(learning_rate=0.01)),  # Fixed-rate baseline.
    ("Adam", lambda: tf.keras.optimizers.Adam(learning_rate=0.001)),  # Adaptive moments.
    ("RMSprop", lambda: tf.keras.optimizers.RMSprop(learning_rate=0.001)),  # Running square average.
    ("Adadelta", lambda: tf.keras.optimizers.Adadelta(learning_rate=1.0)),  # Adaptive update scaling.
    ("Adagrad", lambda: tf.keras.optimizers.Adagrad(learning_rate=0.01)),  # Accumulated gradients.
]  # Finish the optimizer configurations.

results = []  # Collect a summary row for each model-and-optimizer combination.
epoch_rows = []  # Collect per-epoch validation accuracy so convergence can be inspected.
for model_name, builder in [("ANN", build_ann), ("MLP", build_mlp)]:  # Test both architectures.
    tf.keras.utils.set_random_seed(SEED)  # Reproduce the initial weights for this architecture.
    reference_model = builder()  # Create one reference model to record its initial weights.
    initial_weights = reference_model.get_weights()  # Store the same starting weights for all runs.
    del reference_model  # Remove the reference because it is never trained.
    histories = {}  # Save the validation accuracy curves for this architecture.
    for opt_name, make_optimizer in OPTIMIZERS:  # Repeat for all five optimization algorithms.
        tf.keras.utils.set_random_seed(SEED)  # Reset the random ordering for a fair comparison.
        model = builder()  # Make an independent new model for this optimizer.
        model.set_weights(initial_weights)  # Guarantee identical initial model weights.
        optimizer = make_optimizer()  # Change only this optimizer choice across runs.
        model.compile(optimizer=optimizer,  # Attach the optimizer under test.
                      loss="sparse_categorical_crossentropy",  # Use the integer-label loss.
                      metrics=["accuracy"])  # Track classification accuracy.
        start = time.perf_counter()  # Start a wall-clock timer immediately before training.
        history = model.fit(x_train, y_train,  # Train on the fixed training subset.
                            validation_data=(x_val, y_val),  # Validate after every epoch.
                            epochs=EPOCHS,  # Use the same number of epochs every time.
                            batch_size=BATCH_SIZE,  # Use the same mini-batch size every time.
                            shuffle=True,  # Shuffle the training data each epoch.
                            verbose=0)  # Train without verbose progress logs.
        seconds = time.perf_counter() - start  # Calculate the elapsed training seconds.
        test_loss, test_acc = model.evaluate(x_test, y_test, verbose=0)  # Evaluate on unseen test images.
        best_val = max(history.history["val_accuracy"])  # Identify the best validation accuracy.
        histories[opt_name] = history.history["val_accuracy"]  # Store the full accuracy trend.
        results.append({  # Save the numeric outcomes of this run as one table row.
```

```
            "model": model_name,  # Architecture name (ANN or MLP).
            "optimizer": opt_name,  # Optimizer name.
            "learning_rate": round(float(optimizer.learning_rate.numpy()), 6),  # Learning rate used.
            "best_val_accuracy": round(best_val, 4),  # Best validation accuracy over the epochs.
            "test_accuracy": round(test_acc, 4),  # Accuracy on the held-out test subset.
            "test_loss": round(test_loss, 4),  # Loss on the held-out test subset.
            "train_seconds": round(seconds, 2),  # Training time on this machine.
        })  # Finish the result row.
        for epoch_no, acc in enumerate(history.history["val_accuracy"], start=1):  # Walk through epochs.
            epoch_rows.append({"model": model_name, "optimizer": opt_name,  # Identify the run.
                               "epoch": epoch_no, "val_accuracy": round(acc, 4)})  # Save its accuracy.
        print(f"{model_name:3} | {opt_name:8} | val={best_val:.3f} | test={test_acc:.3f} | {seconds:.1f}s")  # Show it.
    plt.figure(figsize=(8, 4.5))  # Start an individual accuracy figure for this architecture.
    for opt_name, val_curve in histories.items():  # Add one line for each optimizer.
        plt.plot(range(1, EPOCHS + 1), val_curve, marker="o", label=opt_name)  # Draw its true history.
    plt.title(f"{model_name}: validation accuracy by optimizer")  # Identify the experiment.
    plt.xlabel("Epoch")  # Label the horizontal axis as epochs.
    plt.ylabel("Validation accuracy")  # Label the vertical axis as validation accuracy.
    plt.xticks(range(1, EPOCHS + 1))  # Show whole-number epoch ticks.
    plt.grid(alpha=0.25)  # Add subtle guidelines for easier comparisons.
    plt.legend()  # Identify each optimizer curve.
    plt.tight_layout()  # Prevent figure labels from being cut off.
    plt.savefig(OUT / f"{model_name.lower()}_validation_accuracy.png", dpi=160)  # Save the graph.
    plt.close()  # Release figure memory before the next model.

with (OUT / "optimizer_results.csv").open("w", newline="", encoding="utf-8") as file:  # Open the CSV.
    writer = csv.DictWriter(file, fieldnames=list(results[0].keys()))  # Use dictionary columns as the table.
    writer.writeheader()  # Write descriptive column names first.
    writer.writerows(results)  # Write every actual measured model result.
with (OUT / "epoch_validation_accuracy.csv").open("w", newline="", encoding="utf-8") as file:  # Second CSV.
    writer = csv.DictWriter(file, fieldnames=list(epoch_rows[0].keys()))  # Columns for the epoch table.
    writer.writeheader()  # Write the column names.
    writer.writerows(epoch_rows)  # Write the per-epoch validation accuracies.
print(f"Results and plots saved to: {OUT.resolve()}")  # Show where to find the files.
```

4. Results (measured by running the code)

All values below were produced by executing the code above; none were typed in or estimated. The same numbers are saved by the program in `optimizer_results.csv` (columns: `model`, `optimizer`, `learning_rate`, `best_val_accuracy`, `test_accuracy`, `test_loss`, `train_seconds`). The best value in each group is shaded green.

Table 1. Student result table (measured)

Model	Optimizer	Best validation acc.	Test acc.	Seconds
ANN	SGD	0.7233	0.7433	1.46
ANN	Adam	0.8883	0.8850	1.59
ANN	RMSprop	0.8833	0.8833	1.65
ANN	Adadelta	0.8833	0.8858	2.03
ANN	Adagrad	0.8283	0.8367	1.71
MLP	SGD	0.7683	0.7667	1.74
MLP	Adam	0.9033	0.8983	2.20
MLP	RMSprop	0.9050	0.8792	2.10
MLP	Adadelta	0.9017	0.8933	1.97
MLP	Adagrad	0.8400	0.8383	1.95

Table 2. Validation accuracy after each epoch (measured)

Model	Optimizer	Epoch 1	Epoch 2	Epoch 3	Epoch 4
ANN	SGD	0.3150	0.5633	0.6883	0.7233
ANN	Adam	0.8267	0.8667	0.8800	0.8883
ANN	RMSprop	0.8250	0.8750	0.8800	0.8833
ANN	Adadelta	0.8317	0.8717	0.8783	0.8833
ANN	Adagrad	0.6967	0.7733	0.8067	0.8283
MLP	SGD	0.3367	0.6133	0.7250	0.7683
MLP	Adam	0.8550	0.8850	0.8950	0.9033
MLP	RMSprop	0.8633	0.8883	0.9033	0.9050
MLP	Adadelta	0.8500	0.8850	0.8933	0.9017
MLP	Adagrad	0.7217	0.7967	0.8267	0.8400

Accuracy comparison graphs

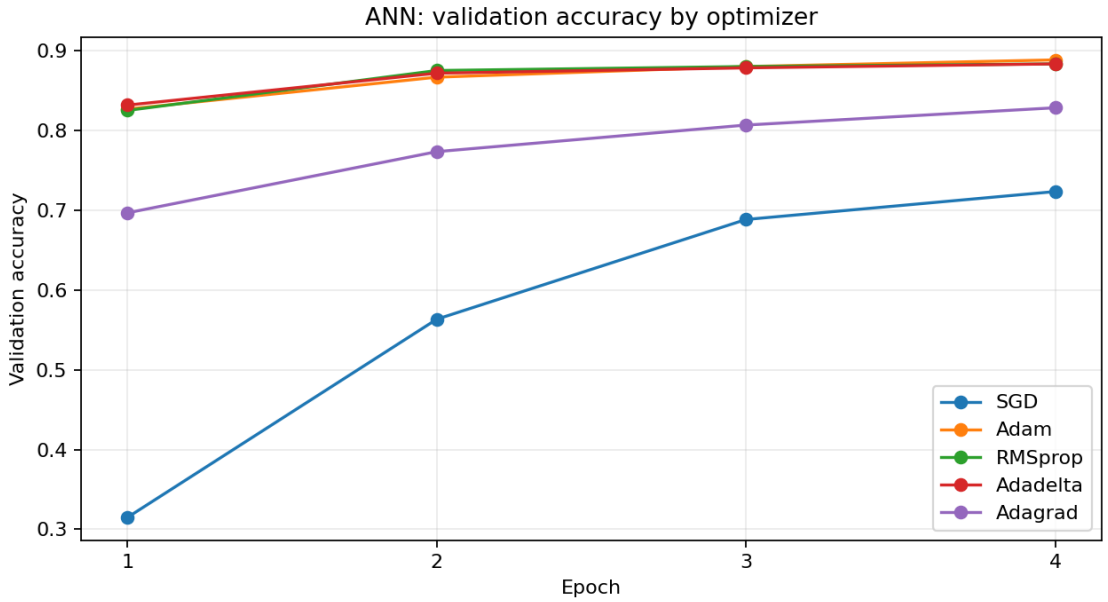


Figure 1. ANN: validation accuracy per epoch for each optimizer (measured).

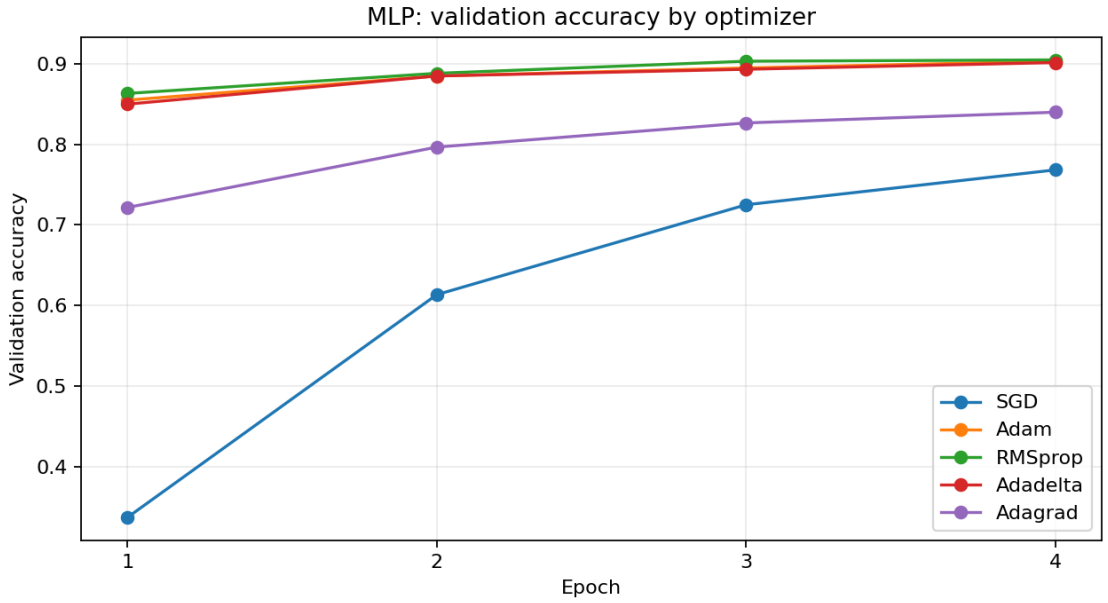


Figure 2. MLP: validation accuracy per epoch for each optimizer (measured).

5. Discussion

Within the ANN, Adam reached the highest best validation accuracy (0.888), with RMSprop and Adadelta tied just behind at 0.883. Within the MLP, RMSprop was highest (0.905), but Adam (0.903) and Adadelta (0.902) differ from it by only one or two of the 600 validation images. SGD (0.723 and 0.768) and Adagrad (0.828 and 0.840) trailed clearly, although each used one fixed learning rate for only four epochs. It is not justified to claim that an adaptive optimizer always improves results: this is a single seed, one small data subset and one untuned learning rate per optimizer, the leaders lie within noise, and the test ranking differs from the validation ranking (for the MLP, Adam has the best test accuracy, 0.898, while RMSprop scores 0.879). A different learning rate could reorder the results. The second hidden layer raised validation accuracy for all five optimizers, for example Adam from 0.888 to 0.903, so extra depth helped here; on the test

set, however, RMSprop slipped slightly (0.883 to 0.879), and the leading optimizer changed between architectures.

What changed, what stayed fixed, and limitations

- **Changed:** only the optimizer object (SGD, Adam, RMSprop, Adadelta, Adagrad) and its learning rate; and, between the two blocks of runs, the architecture (ANN vs MLP).
- **Fixed:** image subsets, pixel scaling, seed, initial weights for each architecture, loss, EPOCHS = 4, BATCH_SIZE = 64, and shuffling seed.
- **Limitations:** one seed per run, only four epochs, 600 validation images (one image is 0.17 percentage points), learning rates not tuned, and training seconds measured on one CPU so they carry timing noise.

Interpretation of the results

Which optimizer is best? By best validation accuracy, Adam for the ANN (0.888) and RMSprop for the MLP (0.905). The top three adaptive optimizers are within about half a percentage point of each other, so none is clearly superior.

Was the comparison controlled? Yes for data, architecture, epochs, batch size, seed and initial weights, which were identical for every run. The learning rate is the one deliberate exception because each optimizer has a different natural scale.

Why are the initial rates different? SGD and Adagrad work with rates around 0.01, Adam and RMSprop with about 0.001, while Adadelta is conventionally started at 1.0 because it rescales its own step. They are illustrative starting points, not tuned values.

ANN vs MLP The MLP scored higher validation accuracy than the ANN with every optimizer (for example SGD 0.723 to 0.768), and its measured training time was similar or slightly higher (1.7 to 2.2 s against 1.5 to 2.0 s; single runs, so only indicative).

6. Reproducibility and environment

Python 3.12, TensorFlow 2.21.0 and Keras 3.15.1 on a CPU-only runtime (one core); random seed 42 is applied before every run. The dataset is the standard public set provided through `tf.keras.datasets`. Because hardware and library versions differ, re-running in Colab can change the *seconds* noticeably and the accuracies slightly; this report quotes only the values from the run shown here.

Files produced by the code: `optimizer_results.csv`, `epoch_validation_accuracy.csv`, and the PNG graphs shown above.

7. References

Keras optimizers reference: https://keras.io/api/optimizers/

TensorFlow Fashion-MNIST tutorial: https://www.tensorflow.org/tutorials/keras/classification

TensorFlow CNN tutorial: https://www.tensorflow.org/tutorials/images/cnn

Google Machine Learning Crash Course (hyperparameters): https://developers.google.com/machine-learning/crash-course/linear-regression/hyperparameters